

33. Morphological Operations Using OpenCV – Morphological Gradient Technique

Aim

To perform morphological gradient on an image using OpenCV.

Procedure

1. Import the required OpenCV and NumPy libraries.
2. Read the input image.
3. Convert the image into grayscale.
4. Create a structuring element (kernel).
5. Apply the morphological gradient operation.
6. Display the original and processed images.
7. Save the output image.

Program

```
import cv2

import numpy as np

# Read the input image

img = cv2.imread("input.jpg")

# Convert to grayscale

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Create structuring element

kernel = np.ones((5, 5), np.uint8)
```

```
# Apply morphological gradient
```

```
gradient = cv2.morphologyEx(gray, cv2.MORPH_GRADIENT, kernel)
```

```
# Display images
```

```
cv2.imshow("Original Image", img)
```

```
cv2.imshow("Morphological Gradient", gradient)
```

```
# Save output image
```

```
cv2.imwrite("gradient_output.jpg", gradient)
```

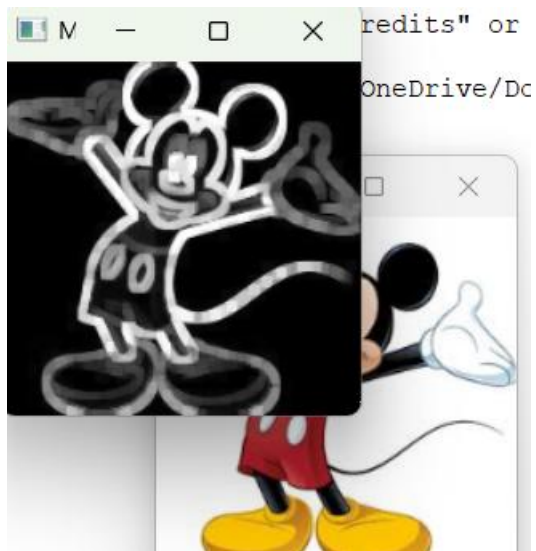
```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

Input



Output



Result

Thus, the morphological gradient operation was successfully performed using OpenCV. It is used to highlight the edges and boundaries of objects in an image.